

EXECUTIVE SUMMARY:

**IMPACT OF 6G NETWORK PERFORMANCE ON
MANUFACTURING EFFICIENCY IN SMART
FACTORIES**

Strategic Technical Brief

Prepared For:

Executive Management Board & Stakeholders

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I. STRATEGIC BUSINESS OVERVIEW

The global smart manufacturing and Industry 4.0 sectors are undergoing a rapid digital transformation. As smart factories increasingly depend on connected machines, automated assembly lines, robotics, and real-time telemetry, network performance becomes an important operational factor in manufacturing efficiency.

This analytical initiative evaluates the relationship between 6G-oriented network performance parameters and factory outcomes, with particular emphasis on **network latency**, **packet loss**, **production speed**, and **quality-control defect rate**. The project moves beyond conventional manufacturing analysis by incorporating communication-network telemetry directly into the efficiency assessment.

The resulting framework provides a data-driven basis for understanding how communication stability can influence production throughput, machine synchronization, and quality outcomes. It also creates a foundation for proactive network monitoring in smart-factory environments.

II. METHODOLOGICAL RIGOR & INNOVATION

The analytical foundation is the **Thales_Group_Manufacturing.csv** dataset, described in the project implementation as containing 100,000 operational telemetry records from advanced smart-factory environments. The analysis was implemented in Python using Pandas, NumPy, Matplotlib, and Seaborn within Google Colab.

The methodology begins with dataset ingestion, structural inspection, missing-value checks, and validation of the core manufacturing and network variables. The principal fields include **Network_Latency_ms**, **Packet_Loss_%**, **Production_Speed_units_per_hr**, **Quality_Control_Defect_Rate_%**, and **Efficiency_Status**.

The project then engineers two dedicated indicators: the **Network Stability Index**, which combines normalized latency and packet-loss penalties, and the **Latency Sensitivity Score**, which relates production speed to observed latency. These indicators extend the analysis beyond individual network measurements and provide a compact view of factory connectivity health and production sensitivity.

III. CRITICAL FINDINGS & KEY PERFORMANCE INDICATORS (KPIs)

1. Network Stability Index

The project combines network latency and packet loss into a normalized inverse-scale indicator. Higher values represent stronger network stability, while increased latency and packet loss reduce the resulting score.

2. Latency Sensitivity Score

The Latency Sensitivity Score is calculated from production speed relative to network latency. It provides an operational indicator of how production performance relates to communication delay.

3. Latency vs Production Speed

The analytical visualization directly compares Network_Latency_ms with Production_Speed_units_per_hr and separates observations by Efficiency_Status. This makes it possible to inspect whether slower communication conditions coincide with lower production speed.

4. Packet Loss vs Quality Defect Rate

A second visualization cross-analyzes Packet_Loss_% against Quality_Control_Defect_Rate_%. This relationship is used to examine whether communication instability is associated with higher quality-control defects.

5. Operational Thresholds

The project framework identifies latency below 15 ms as an optimal high-speed operating condition, 15–45 ms as an at-risk latency range, and latency above 45 ms combined with high packet loss as a critical-degradation condition. These thresholds are used as analytical classification criteria in the report.

IV. OPERATIONAL IMPLEMENTATION & REAL-TIME MONITORING

To convert the analytical findings into an operational tool, the project specifies a live Streamlit web application titled “**Impact of 6G Network Performance on Manufacturing Efficiency**”. The application is designed to give engineering and factory stakeholders a real-time view of network and production indicators.

The dashboard provides filtering by **Efficiency Status** and a configurable maximum latency threshold. It also presents KPI cards for total records, average latency, average Network Stability Index, and average quality-control defect rate.

Two principal analytical views are embedded in the dashboard: **Latency vs Production Speed** and **Packet Loss vs Defect Rate**. Together, these views connect network telemetry to manufacturing performance and support rapid identification of potentially degraded operating conditions.

The processed dataset containing the engineered KPIs is saved as **Thales_Group_Manufacturing_Processed.csv** for use by the Streamlit application. This establishes a repeatable pipeline from raw telemetry to an interactive monitoring interface.

V. MANDATED STRATEGIC DIRECTIVES

Automated Network Threshold Alerts: Integrate latency and packet-loss thresholds into factory management systems so that significant network degradation can be detected before it affects production-line continuity.

Preemptive Edge-Routing Optimization: When the system identifies an at-risk latency condition, engineering teams can review edge-routing and local connectivity configurations to reduce communication delays.

Critical Network Failover: For conditions characterized by high latency and high packet loss, implement automated failover procedures to maintain communication reliability for time-sensitive manufacturing operations.

Continuous KPI Monitoring: Use the Streamlit dashboard as a recurring operational monitoring interface, with particular attention to Network Stability Index, average latency, and quality-control defect rate.

Conclusion: The project establishes a structured data-science framework for examining how 6G network performance can affect smart-factory manufacturing efficiency. By combining network telemetry, production metrics, engineered KPIs, visual analysis, and an interactive dashboard, the framework supports proactive monitoring and data-driven industrial decision-making.